Containerization-and-DevOps

🚀 Class Task

🛠️ Tools Used

- Docker
- Portainer

⚙️ Steps

Create Docker Volume

```
docker volume create portainer_data
```

Run Portainer Container

```
docker run -d -p 8000:8000 -p 9443:9443 \
--name portainer \
--restart=always \
-v /var/run/docker.sock:/var/run/docker.sock \
-v portainer_data:/data \
portainer/portainer-ce:lts
```

Verify Running Container

```
docker ps
```

Access Portainer

Open in browser:

```
https://localhost:9443
```

Initial Setup

- Create admin username & password
- Select **Docker environment (local)**
- Connect to Docker

⚠️ Notes

- Portainer image is pulled automatically if not present
- Docker socket is mounted for container management access
- Volume ensures persistent data storage
- First-time setup requires creating admin credentials
- Uses HTTPS (port 9443) for secure access 🔐

✅ Outcome

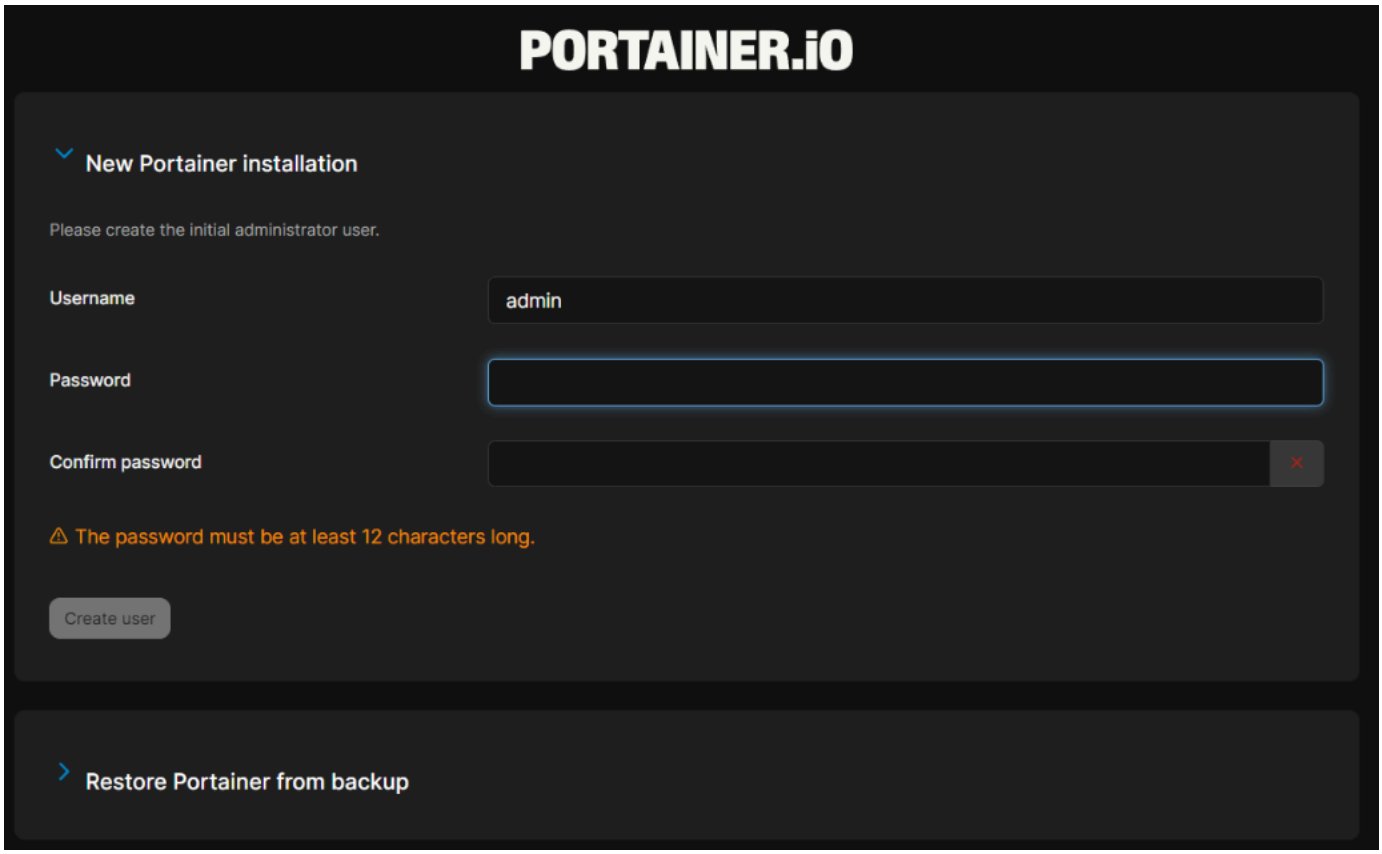
- Portainer container successfully deployed
- Web UI accessed via browser
- Docker environment connected
- Containers, volumes, and images visible through GUI
- Simplified Docker management using Portainer

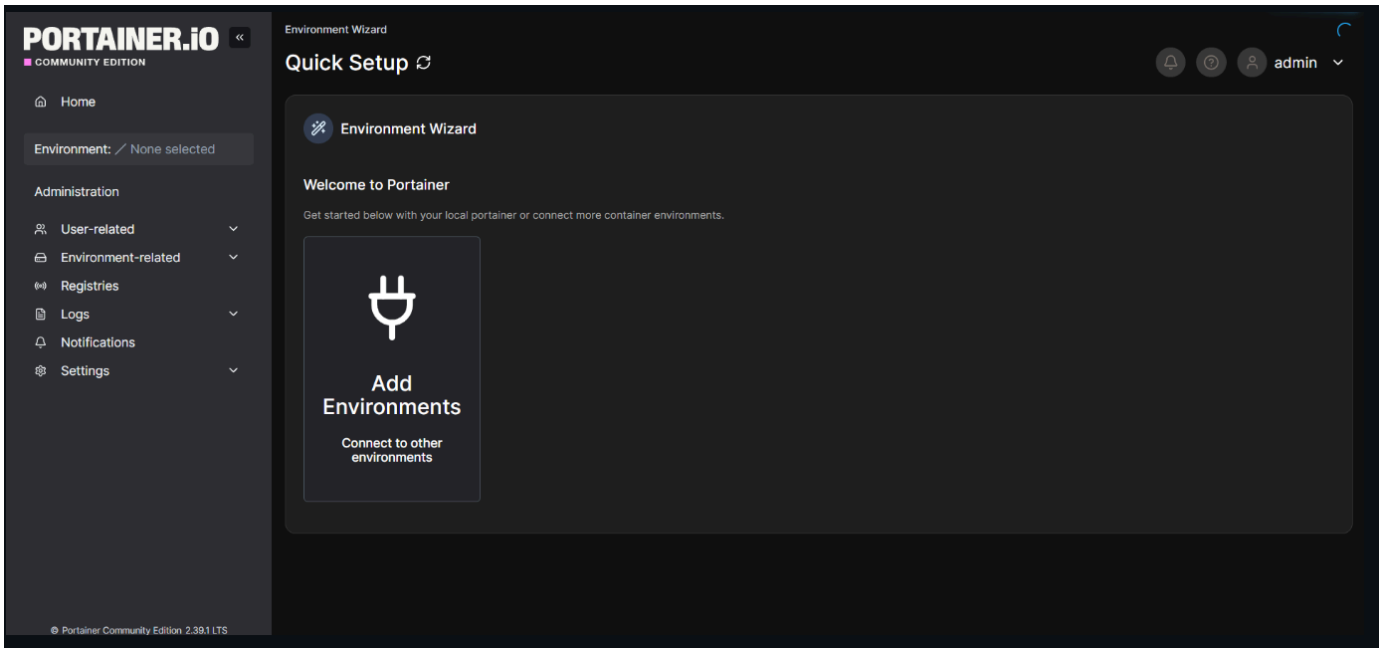
📸 Screenshots

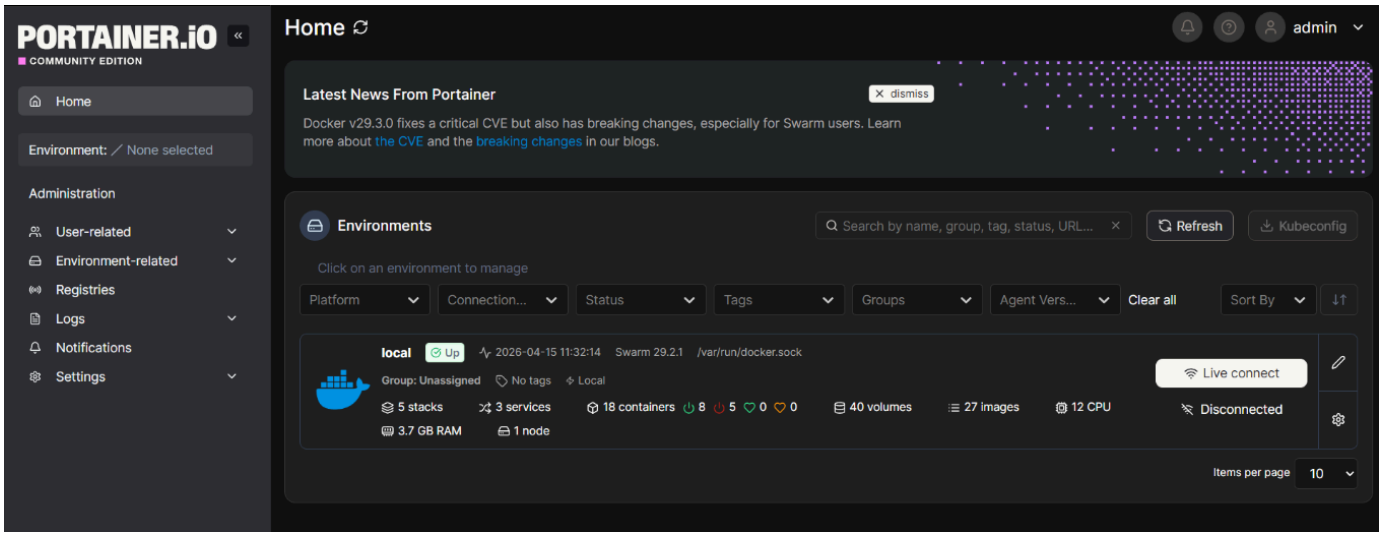
```
sristee@Sristee:/mnt/c/Users/srist$ docker volume create portainer_data
portainer_data
```

```
sristee@Sristee:/mnt/c/Users/srist$ docker run -d -p 8000:8000 -p 9443:9443 --name portainer --restart=always -v /var/run/docker.sock
:/var/run/docker.sock -v portainer_data:/data portainer/portainer-ce:lts
Unable to find image 'portainer/portainer-ce:lts' locally
lts: Pulling from portainer/portainer-ce
4f4fb700ef54: Pull complete
d1e89ff9b379: Pull complete
9ab41045230a: Pull complete
2ff62971c2fb: Pull complete
1c4fea3af7a0: Pull complete
70636407b8c2: Pull complete
c7e34731674e: Pull complete
224552a17e45: Pull complete
Digest: sha256:1ae8e65d50ca5498cb2c33e617495a1e3ef245b0d2392b4a44c70ae09b822891
Status: Downloaded newer image for portainer/portainer-ce:lts
ca750feb19c2484a2a11177fbe8a1ccd07a4cffc8482d341c14baff8a0c7c565
```

```
sristee@Sristee:/mnt/c/Users/srist$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS                          PORTS
                                                                                   NAMES
ca750feb19c2   portainer/portainer-ce:lts     "/portainer"             21 seconds ago   Up 20 seconds                   0.0.0.0:800
0->8000/tcp, [::]:8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp       portainer
77d0f945b7e6   ghcr.io/k3d-io/k3d-tools:5.8.3   "/app/k3d-tools noop"    4 hours ago      Up 4 hours
                                                                                   k3d-mycluster-tools
2bce202382ca   ghcr.io/k3d-io/k3d-proxy:5.8.3   "/bin/sh -c nginx-pr…"   4 hours ago      Up 4 hours                      0.0.0.0:355
85->6443/tcp                                                                       k3d-mycluster-serverlb
7f05c22cd54a   rancher/k3s:v1.31.5-k3s1         "/bin/k3d-entrypoint…"   4 hours ago      Up 4 hours
                                                                                   k3d-mycluster-server-0
3a6ec2eecb2d   postgres:13                      "docker-entrypoint.s…"   6 hours ago      Up 6 hours                      5432/tcp
                                                                                   sonar-db
7284201b6163   sonarqube:lts-community          "/opt/sonarqube/dock…"   6 hours ago      Up 6 hours                      0.0.0.0:900
0->9000/tcp, [::]:9000->9000/tcp                                                   sonarqube
f25c36a0b318   ubuntu-server                    "/usr/sbin/sshd -D"      6 hours ago      Up 6 hours                      0.0.0.0:822
1->8221/tcp, [::]:8221->8221/tcp, 0.0.0.0:2222->22/tcp, [::]:2222->22/tcp           ssh-test-server
69c176800b45   jenkins/jenkins:lts              "/usr/bin/tini -- /u…"   8 hours ago      Up 8 hours                      0.0.0.0:808
0->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   jenkins
762697afd308   container-app-backend            "docker-entrypoint.s…"   4 weeks ago      Restarting (1) 28 seconds ago
                                                                                   node_backend
f1f291734380   container-app-database           "docker-entrypoint.s…"   4 weeks ago      Up 8 hours (healthy)
                                                                                   postgres_db
```




PORTAINER.io
COMMUNITY EDITION
Home
Environment: None selected
Administration
User-related
Environment-related
Registries
Logs
Notifications
Settings
© Portainer Community Edition 2.39.1 LTS
Environment Wizard
Quick Setup
admin
Environment Wizard
Welcome to Portainer
Get started below with your local portainer or connect more container environments.
Add Environments
Connect to other environments


PORTAINER.io
COMMUNITY EDITION
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admin
Latest News From Portainer
dismiss
Docker v29.3.0 fixes a critical CVE but also has breaking changes, especially for Swarm users. Learn more about the CVE and the breaking changes in our blogs.
Environments
Search by name, group, tag, status, URL...
Refresh
Kubeconfig
Click on an environment to manage
Platform
Connection...
Status
Tags
Groups
Agent Vers...
Clear all
Sort By
local
Up
2026-04-15 11:32:14
Swarm 29.2.1
/var/run/docker.sock
Group: Unassigned
No tags
Local
5 stacks
3 services
18 containers
8
5
0
0
40 volumes
27 images
12 CPU
3.7 GB RAM
1 node
Live connect
Disconnected
Items per page
10